NOETICA MASTER BLUEPRINT

PART 16

Database Architecture & Data Model

Purpose

The Database Architecture is the memory layer of Noetica.

Every capability within Noetica depends on the database.

Without the database there is:

- No discovery tracking
- No knowledge graph
- No personalization
- No forecasting
- No historical intelligence

The database therefore serves as the foundation upon which all higher intelligence layers operate.

Core Philosophy

Most research systems store:

Paper

↓

Metadata

↓

Search

Noetica stores:

Discovery

↓

Relationships

↓

Knowledge Evolution

↓

Intelligence

Database Evolution

V1

SQLite

Purpose:

Validate the platform.

V2

PostgreSQL

Purpose:

Multi-user scientific intelligence platform.

V3+

PostgreSQL

+

Neo4j

+

Vector Database

Purpose:

Global Scientific Intelligence Network.

Primary Database Principles

1. Discoveries are primary entities.
2. Papers are evidence objects.
3. Knowledge is represented through relationships.
4. Historical information is never destroyed.
5. Every score must be explainable.
6. Forecasts must be reproducible.
7. Schema must support future unknown fields.

Core Database Domains

The database is divided into:

Users

Discoveries

Knowledge Graph

Researchers

Organizations

Patents

Grants

Datasets

Reports

Forecasts

Feedback

Alerts

Taxonomy

Discovery Domain

Table: discoveries

Purpose:

Store discovery clusters.

Fields

discovery_id

title

summary

description

field_id

subfield_id

lifecycle_stage

first_seen_date

last_seen_date

status

created_at

updated_at

Example

Discovery:

CRISPR

This discovery becomes the primary object.

Not individual papers.

Discovery Evidence Domain

Table: discovery_evidence

Purpose:

Attach evidence to discoveries.

Fields

evidence_id

discovery_id

source_type

source_id

source_name

publication_date

confidence_score

Examples

Paper

Patent

Grant

Conference

Dataset

Open Source Project

Government Program

Discovery Lifecycle Domain

Table: discovery_lifecycle

Purpose:

Track evolution.

Fields

lifecycle_id

discovery_id

stage

entered_stage_date

confidence

Stages

Speculative

Emerging

Growing

Breakthrough

Established

Foundational

Civilizational

Historical

Discovery Scores Domain

Table: discovery_scores

Purpose:

Store significance metrics.

Fields

score_id

discovery_id

scientific_significance

technological_significance

civilizational_significance

future_potential

momentum_score

confidence_score

last_updated

Scientific Score Breakdown

Table: score_components

Fields

component_id

discovery_id

evidence_strength

novelty

replication

cross_field_influence

knowledge_connectivity

citation_growth

funding_momentum

practical_impact

Purpose

Ranking transparency.

Every score must be explainable.

Knowledge Graph Domain

Table: knowledge_nodes

Purpose:

Store graph nodes.

Fields

node_id

node_name

node_type

field_id

description

importance_score

created_at

Node Types

Discovery

Field

Subfield

Researcher

Organization

Patent

Grant

Dataset

Technology

Theory

Scientific Method

Country

Government Program

Knowledge Relationships

Table: knowledge_edges

Purpose:

Store graph relationships.

Fields

edge_id

source_node

target_node

relationship_type

weight

confidence

created_at

Relationship Types

influences

depends_on

derived_from

enables

contradicts

supports

accelerated_by

commercialized_by

funded_by

implemented_by

competes_with

Taxonomy Domain

Table: fields

Purpose:

Store major knowledge domains.

Examples

Physics

Mathematics

Biology

Economics

Philosophy

Artificial Intelligence

Fields

field_id

name

description

created_at

Subfield Domain

Table: subfields

Purpose:

Store evolving subfields.

Fields

subfield_id

field_id

name

description

growth_rate

created_at

Important

Fields are not hardcoded forever.

Taxonomy evolves automatically.

Emerging Fields Domain

Table: emerging_fields

Purpose:

Track new disciplines.

Fields

emerging_field_id

name

origin_fields

growth_score

confidence

first_detected

status

Examples

AI Drug Discovery

Quantum Biology

Scientific Foundation Models

Autonomous Science

Researcher Domain

Table: researchers

Fields

researcher_id

name

organization_id

primary_field

secondary_fields

influence_score

research_momentum

Researcher Contributions

Table: researcher_discoveries

Purpose:

Connect researchers to discoveries.

Fields

researcher_id

discovery_id

role

contribution_score

Organization Domain

Table: organizations

Fields

organization_id

name

organization_type

country

influence_score

research_output

Organization Types

University

Research Institute

Company

Government Agency

Think Tank

Nonprofit

Patent Intelligence Domain

Table: patents

Fields

patent_id

title

organization_id

technology_area

filing_date

status

importance_score

Grant Intelligence Domain

Table: grants

Fields

grant_id

organization_id

funding_body

amount

field

start_date

end_date

importance_score

Dataset Intelligence Domain

Table: datasets

Fields

dataset_id

name

field

description

usage_score

importance_score

Examples

Human Genome

TCGA

ImageNet

AlphaFold DB

Protein Data Bank

Open Source Intelligence Domain

Table: open_source_projects

Fields

project_id

name

repository_url

organization_id

stars

forks

contributors

adoption_score

Forecast Domain

Table: forecasts

Purpose:

Store future predictions.

Fields

forecast_id

entity_type

entity_id

forecast_type

probability

confidence

generated_at

Forecast Types

Future Significance

Field Growth

Technology Adoption

Discovery Impact

Scientific Consensus Domain

Table: consensus_scores

Purpose:

Track agreement levels.

Fields

consensus_id

discovery_id

consensus_score

supporting_evidence

conflicting_evidence

updated_at

Scientific Debate Domain

Table: debate_scores

Purpose:

Track disagreement.

Fields

debate_id

discovery_id

debate_score

active_controversies

updated_at

User Domain

Table: users

Fields

user_id

email

username

expertise_level

reading_time

report_frequency

created_at

User Interests

Table: user_interests

Fields

user_id

field_id

interest_weight

User Feedback

Table: user_feedback

Fields

feedback_id

user_id

discovery_id

feedback_type

timestamp

Feedback Types

Very Useful

Useful

Neutral

Not Useful

Watchlists

Table: watchlists

Fields

watchlist_id

user_id

entity_type

entity_id

created_at

Alerts

Table: alerts

Fields

alert_id

user_id

alert_type

trigger_condition

is_active

created_at

Reports Domain

Table: reports

Purpose:

Store compressed intelligence outputs.

Fields

report_id

report_type

generation_date

summary

top_discoveries

top_fields

Report Types

Weekly

Monthly

Annual

Important

Daily reports are not permanently archived.

Only:

Weekly

Monthly

Annual

reports are preserved.

Intelligence Audit Layer

Table: audit_logs

Purpose:

Track major system decisions.

Fields

audit_id

entity_type

entity_id

action

reason

timestamp

Purpose

Transparency

Reproducibility

Debugging

Governance

Future Database Evolution

V1

SQLite

↓

V2

PostgreSQL

↓

V3

PostgreSQL

+

Neo4j

↓

V4

Distributed Knowledge Infrastructure

Knowledge Lake

Vector Search

Global Knowledge Graph

End State

The Noetica database is not a paper database.

It is a structured representation of human knowledge.

Its purpose is to preserve discoveries, relationships, significance, history, forecasts, and intelligence in a form that can be explored, analyzed, and continuously expanded as human knowledge evolves.